Lab08: Security and Firewall Management

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Author

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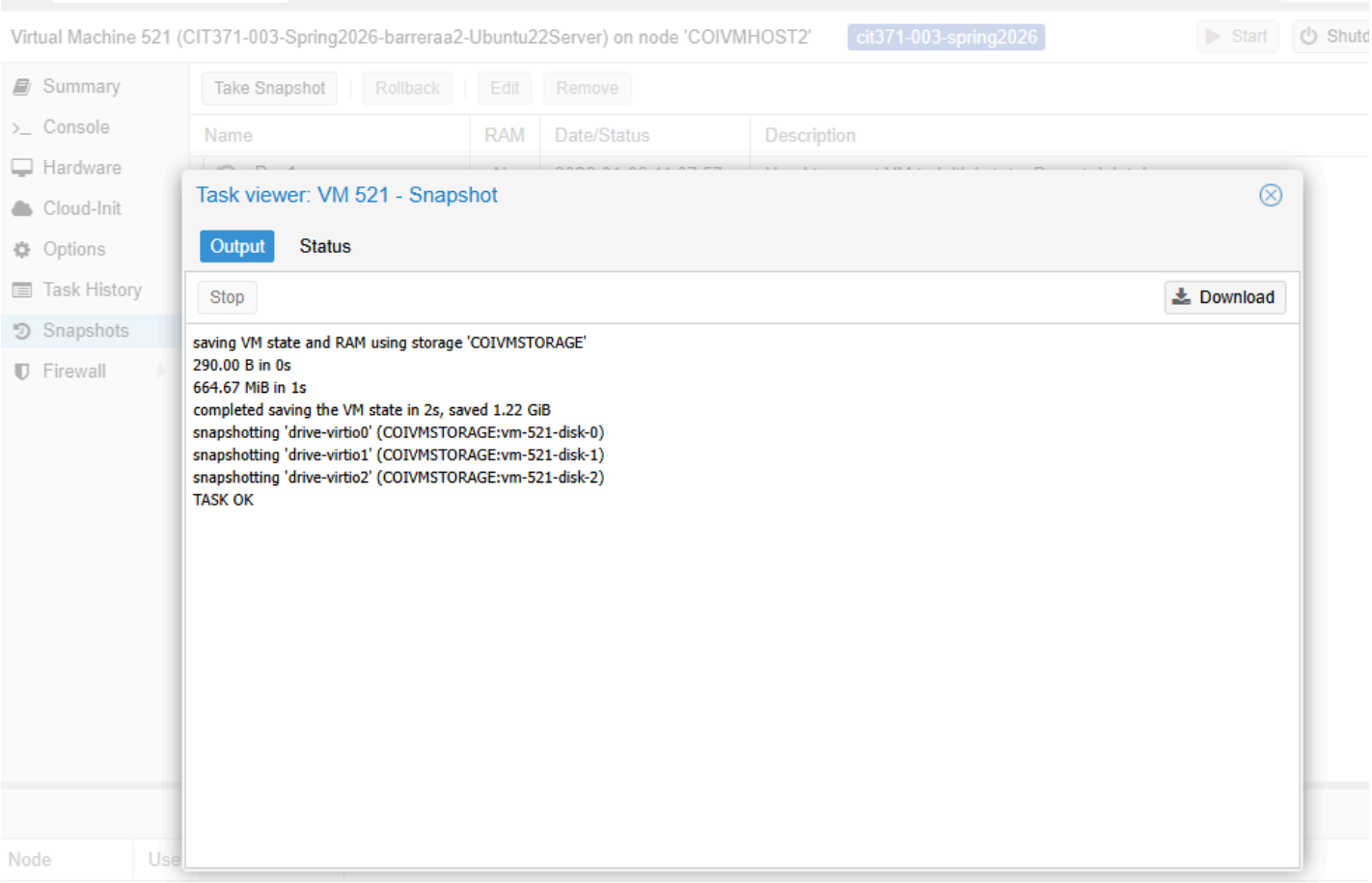
VM Snapshot(s)



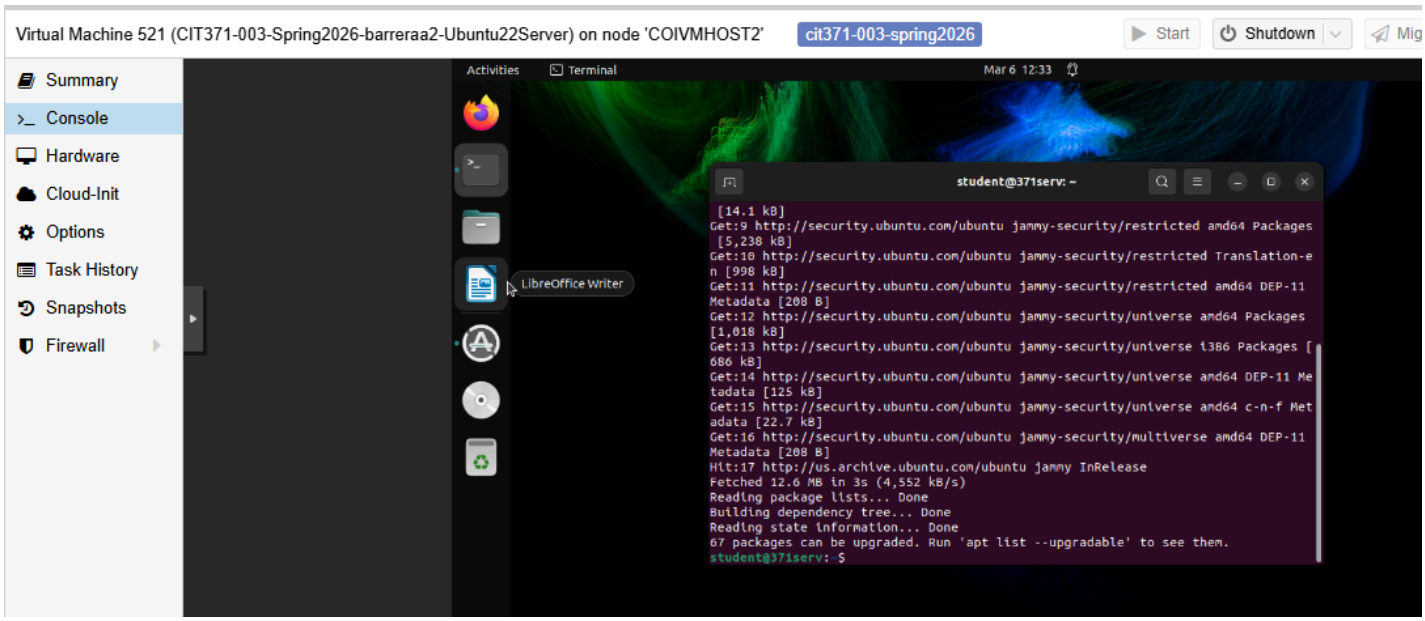
Overview

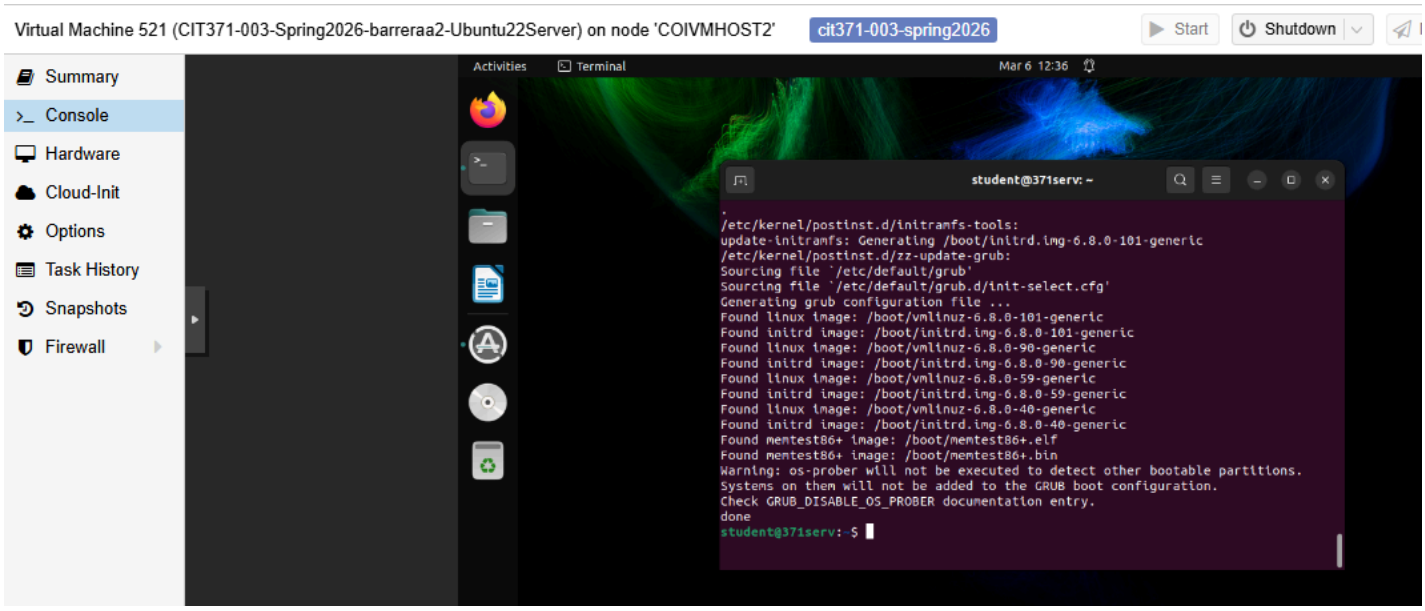
In this lab we will learn some techniques that will help the security of our Ubuntu Server. We will be using commands to update and upgrade our server as well are reveal the accounts that has no password and delete those account. Also, we will working with the GPG and disabling passwords caching for the GPG agent. We will be disabling accounts that has root access for SSH. Ubuntu Firewalls will be configured and enabling logging on them and blocking connections and ports.

PART 2:

We will start with Update and Upgrade our Ubuntu Server with this commands:

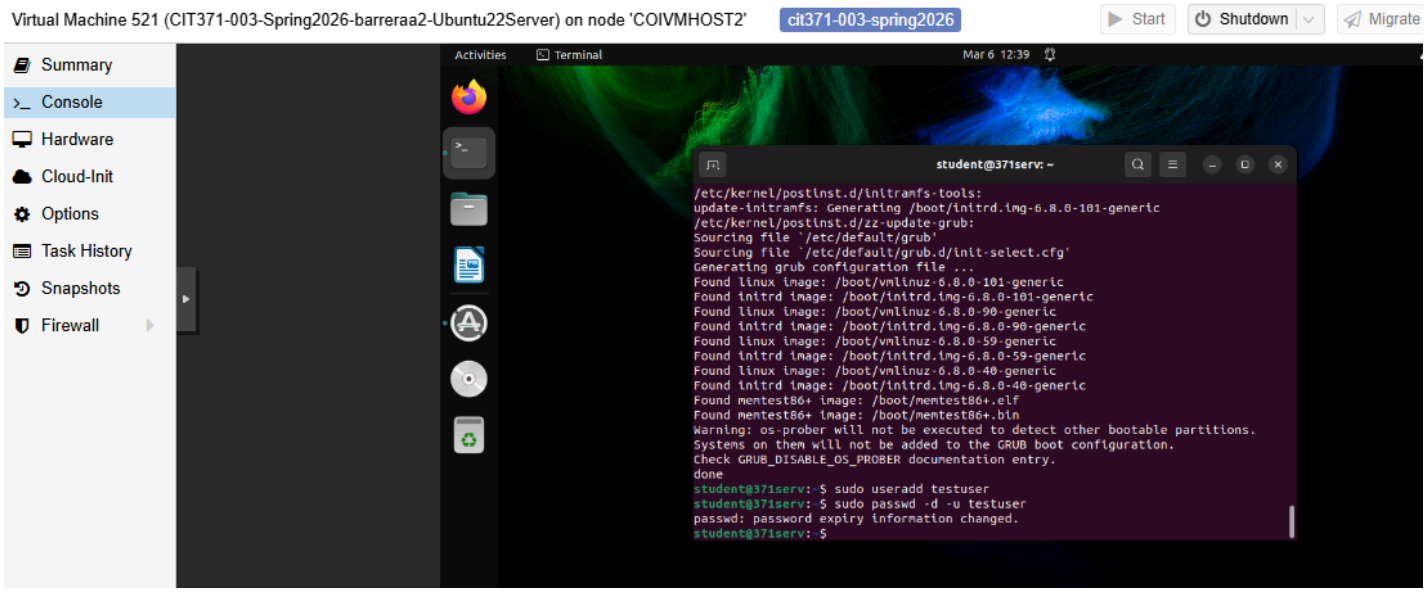
```
sudo useradd testuser
sudo apt upgrade -y
```





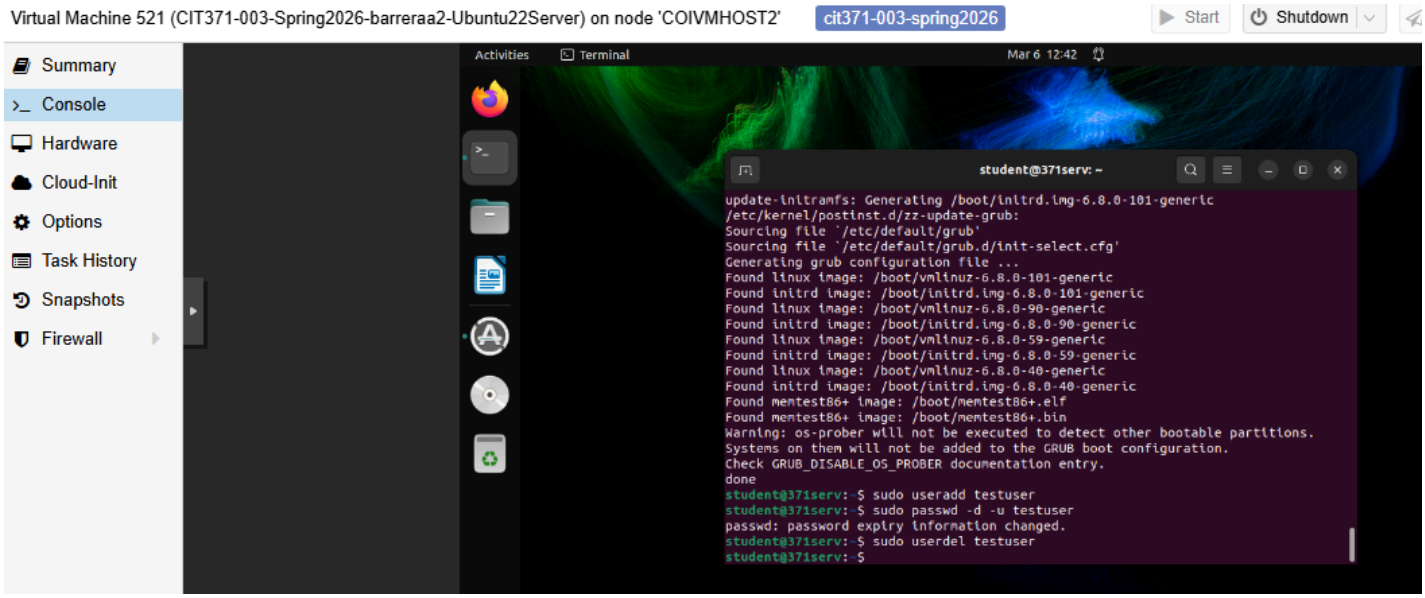
Now that we have patched our Ubuntu server for security, we will need to create a password-less user for testing using this commands:

```
sudo useradd testuser
sudo passwd -d -u testuser
```



Now we will use this command to make sure that we see that this username does not have a password with this command:

```
sudo userdel testuser
```

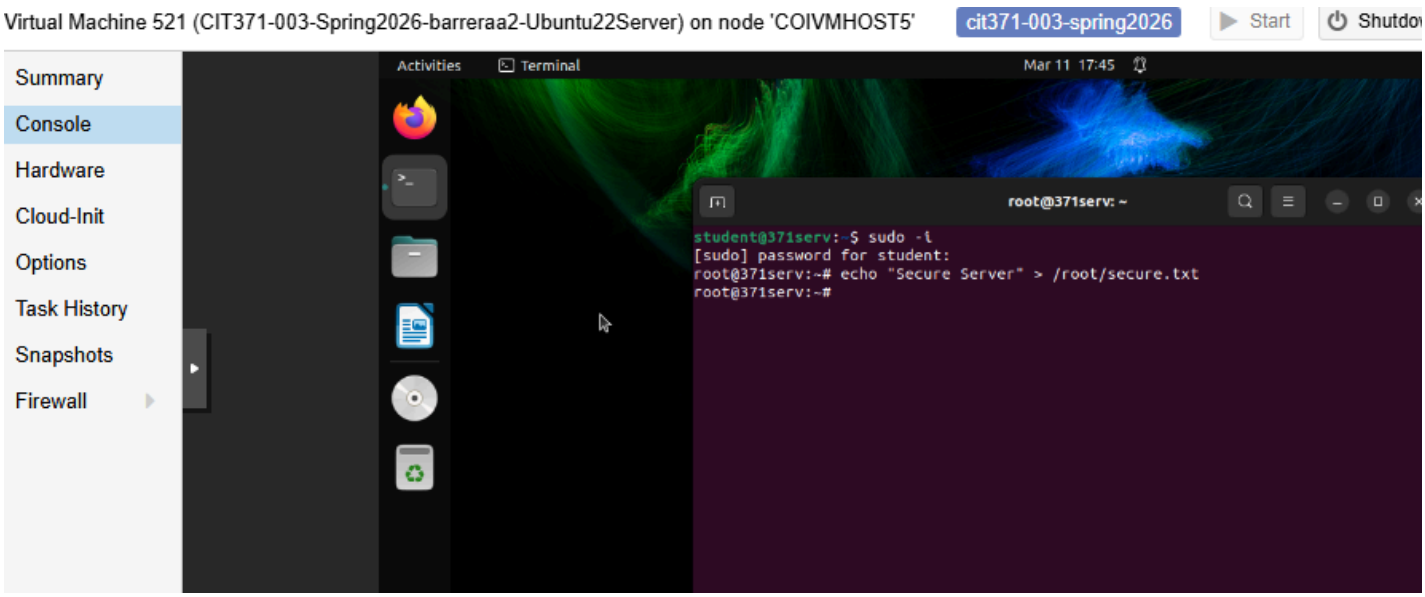


After the delete test account we will now do the GPG encryption task. First we will need to switch to root using this command:

```
sudo -i
```

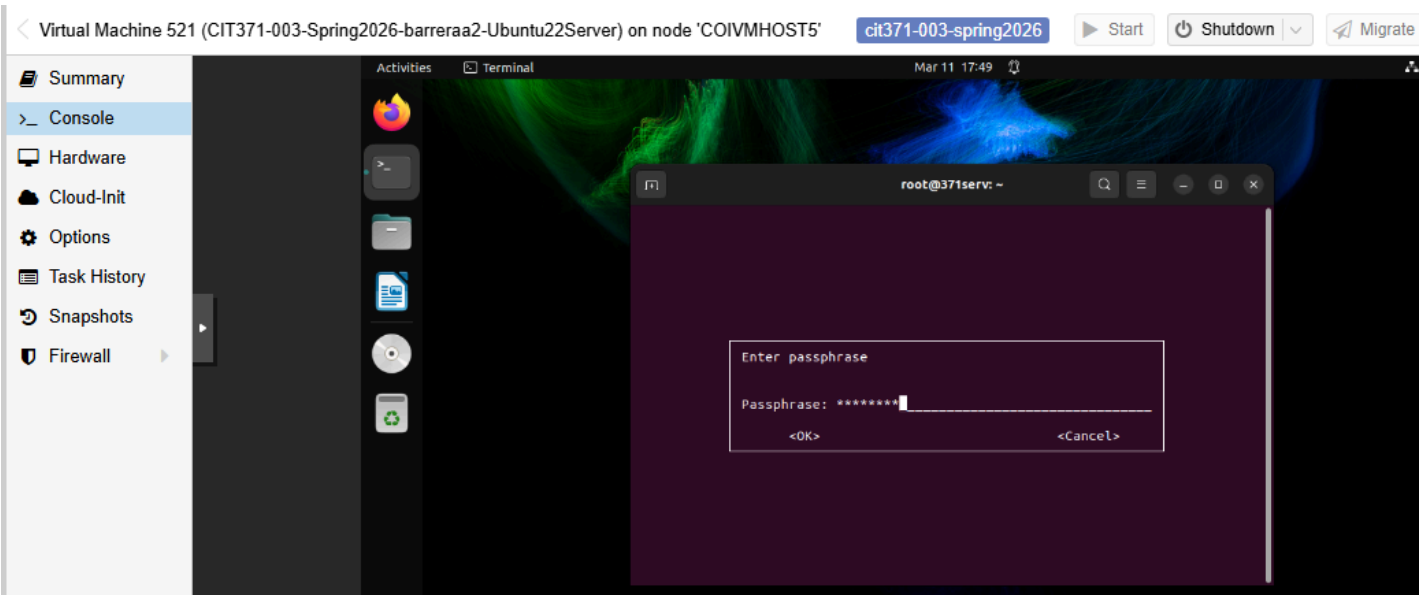
Then we will create the file using this command:

```
echo "Secure Server" > /root/secure.txt
```



Then we will Encrypt the file with a passphrase using this code - then the passphrase will be Passw0rd whenever it asked for it.

```
gpg -c /root/secure.txt
```

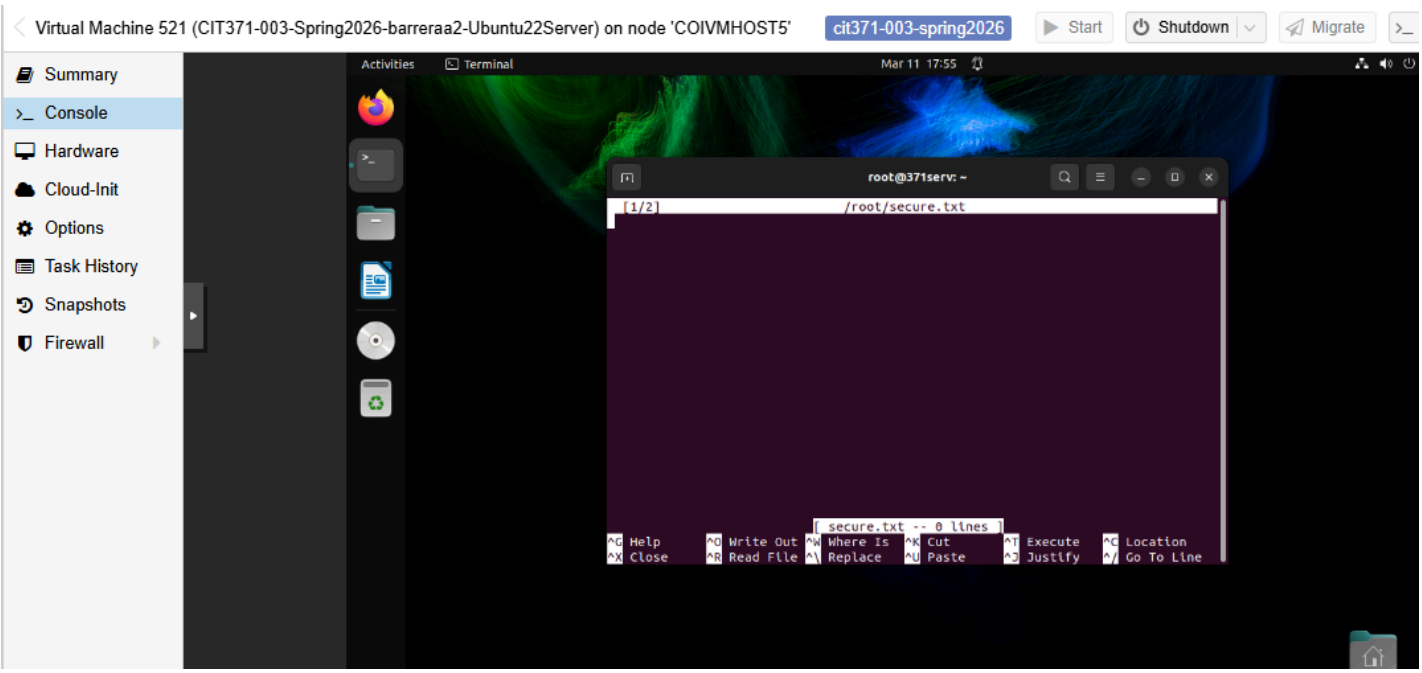


After we have encrypted the file we will remove the encryption using this command:

```
rm /root/secure.txt
```

Then we will make sure we open it in nano to make sure that it shows unreadable encrypted data with this commnad:

```
nano /root/secure.txt gpg
```

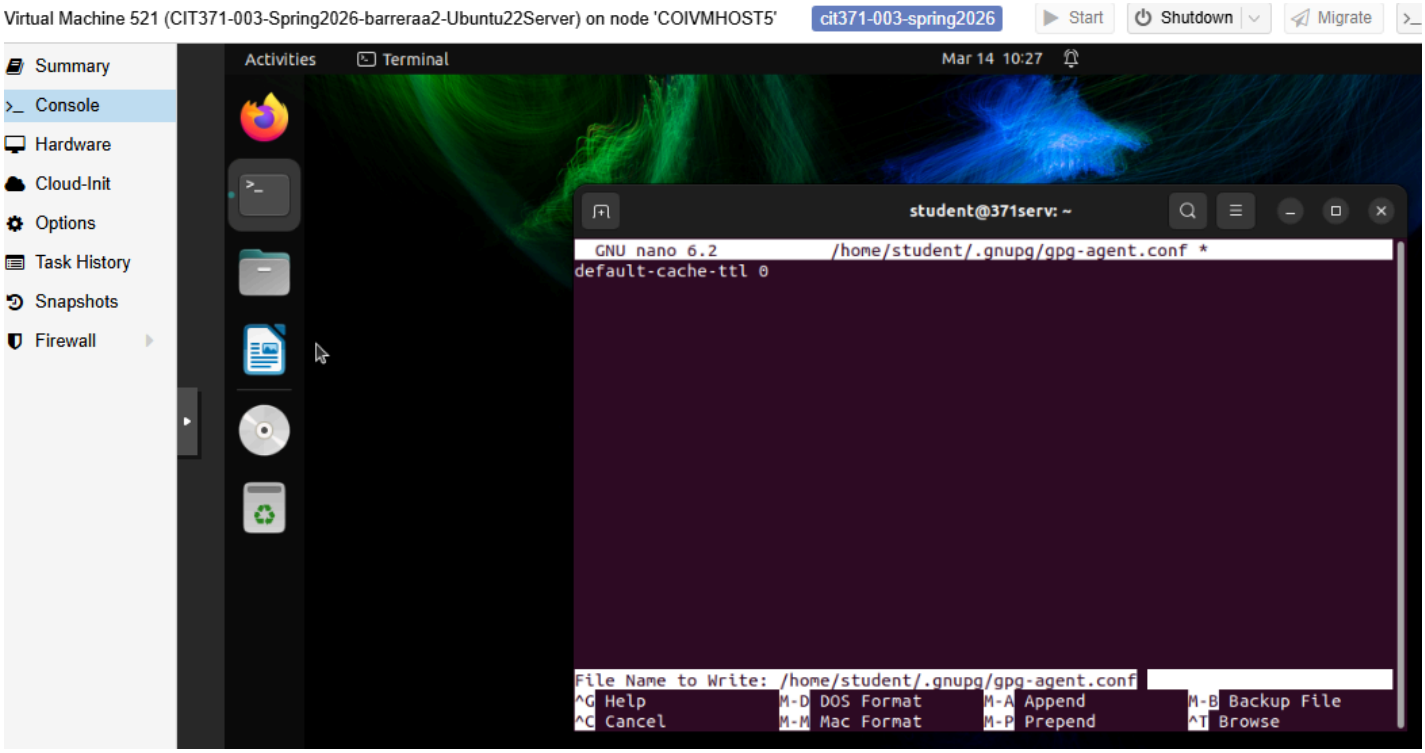


Now we will just exit from the root account by typing this command:

```
exit
```

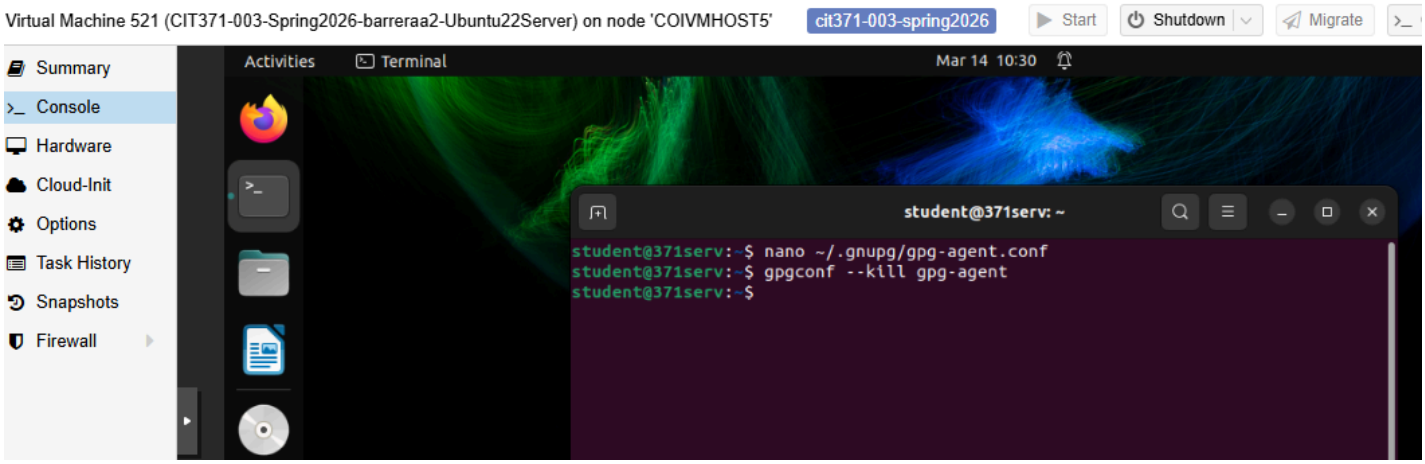
After getting out of root, we will now disable password caching for the GPG agent in the student's home directory using this command:

```
nano ~/.gnupg/gpg-agent.conf
default-cache-ttl 0
```



Then we will reload the agent to make the changes applicable with this command:

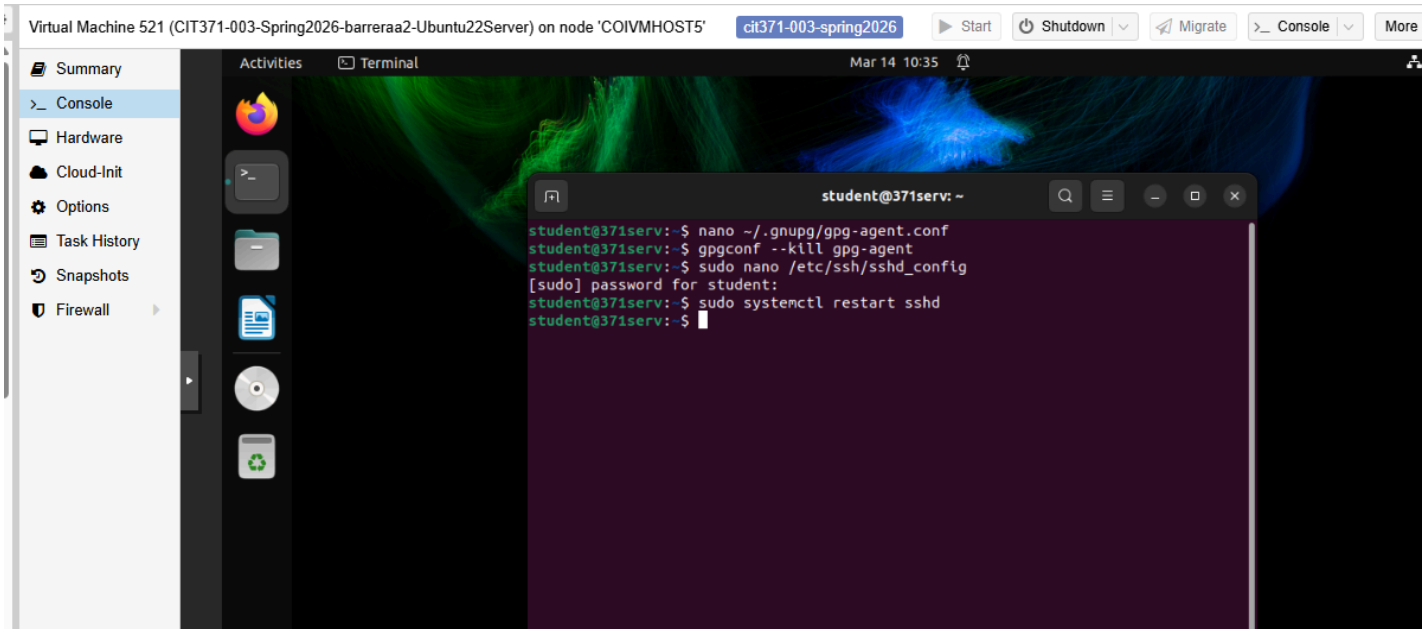
```
gpgconf --kill gpg-agent
```



We will now disable over SSH the root login. This few commands we will put them one by one:

```
sudo nano /etc/ssh/sshd_config
permitrootlogin no
sudo systemctl restart sshd
```

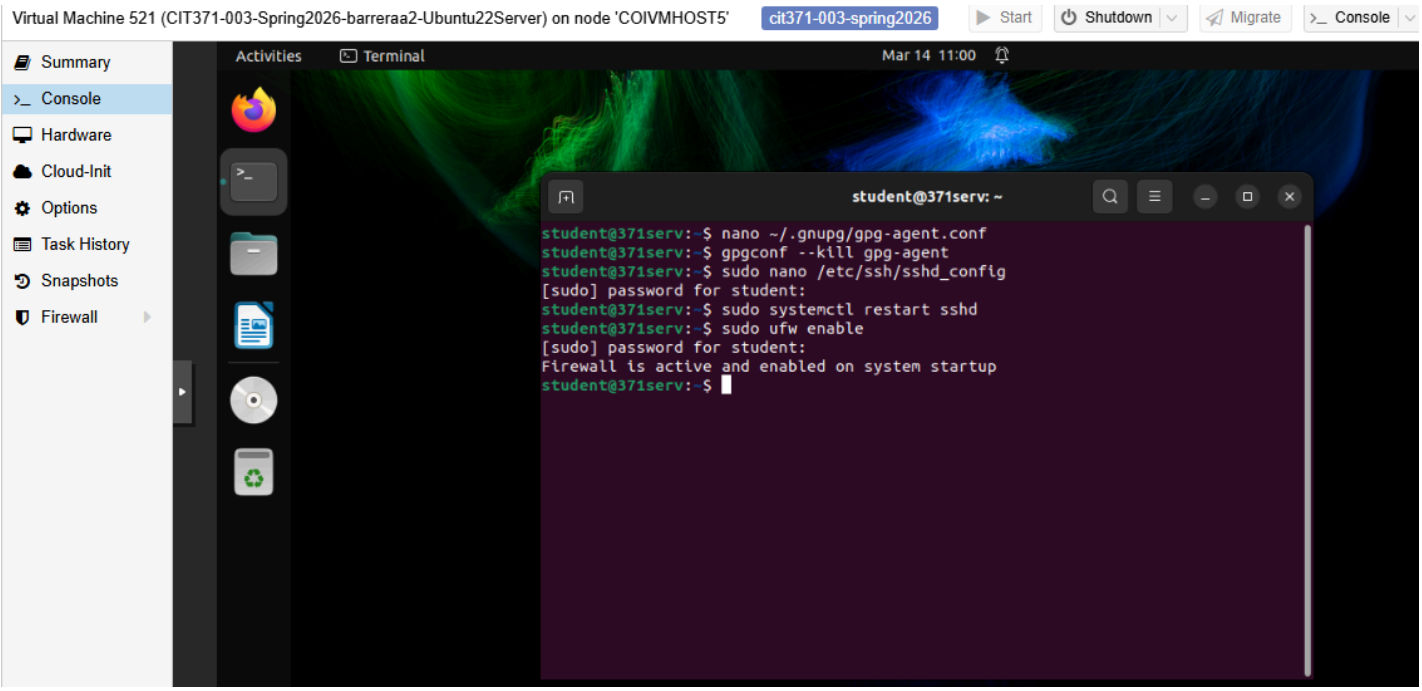
In the last command we want to make sure we restart and we do not need to test login because no root password was set during our installation.



PART 3:

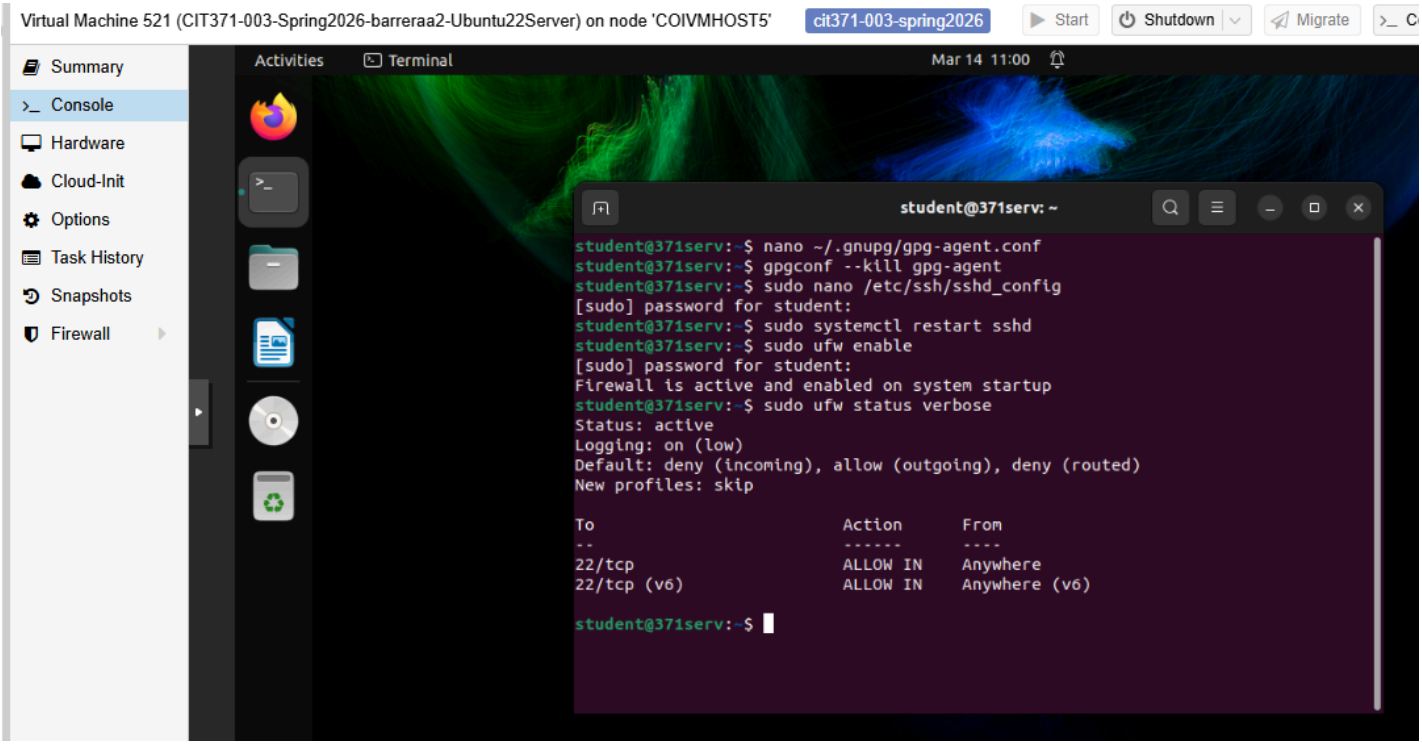
In this part we will enable the UFW firewall with this command:

```
sudo ufw enable
```



Then we will use the verbose option and list all the firewalls current list with this tatus command:

```
sudo ufw status verbose
```



Now we will enable the UFW logging with this command:

```
sudo ufw logging on
```

then we will deny access to port 23 which is telnet with this command:

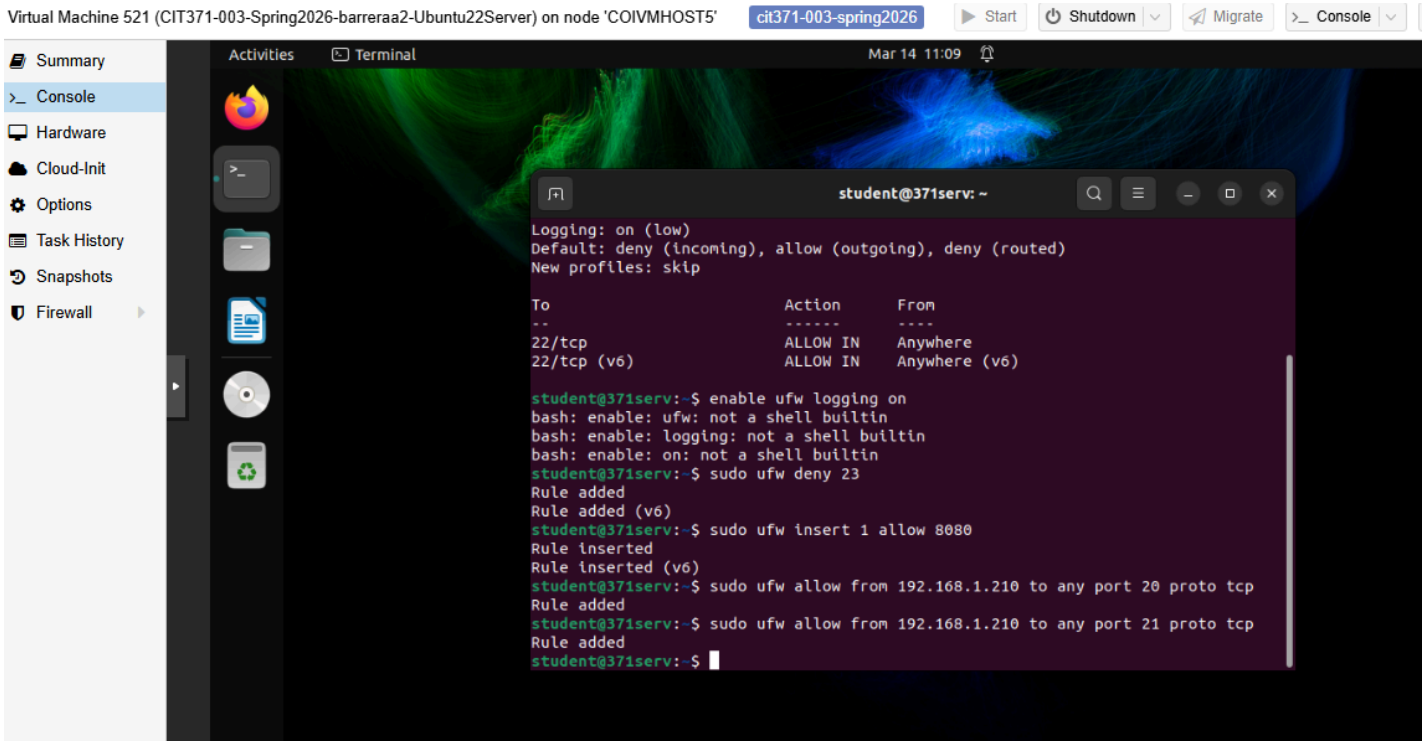
```
sudo ufw deny 23
```

Then we will make some rules, we will make this rule the number 1 and it will be to allow 8080.

```
sudo ufw insert 1 allow 8080
```

Then we will allow ports 20 and 21 TCP from our Ubuntu client with these commands:

```
sudo ufw allow from 192.168.1.210 to any port 20 proto tcp
sudo ufw allow from 192.168.1.210 to any port 21 proto tcp
```



References

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